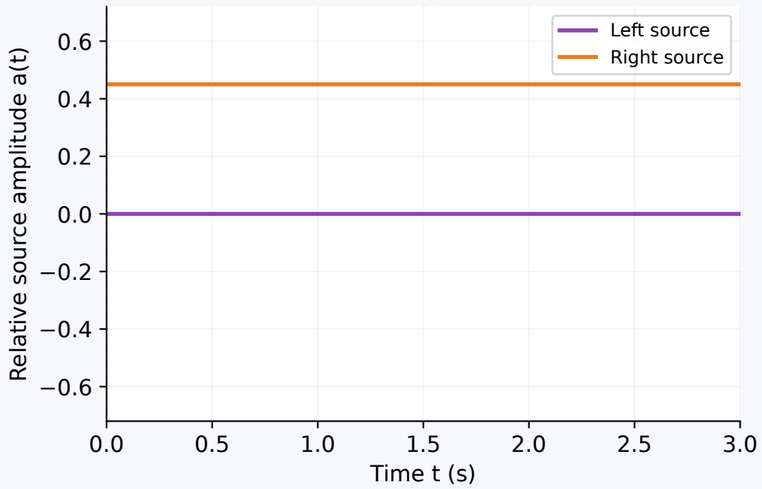
MAGNETIC INPUT → FLUID FLOW → PARTICLE RESPONSE

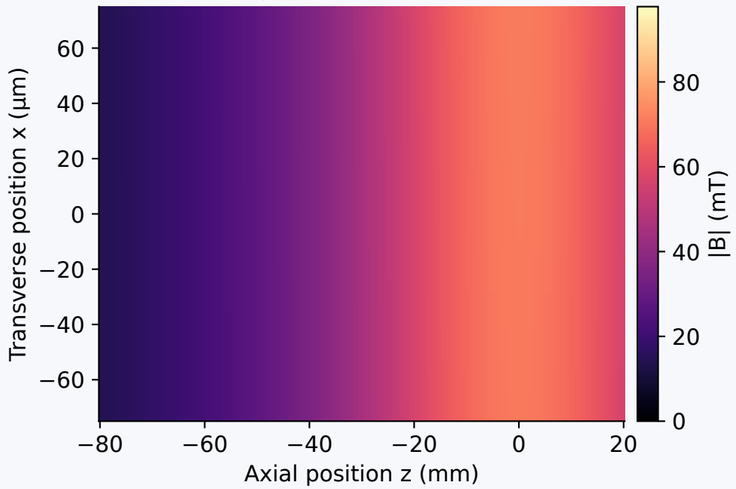
Constant right source

$a_L = 0, \quad a_R = 0.45$

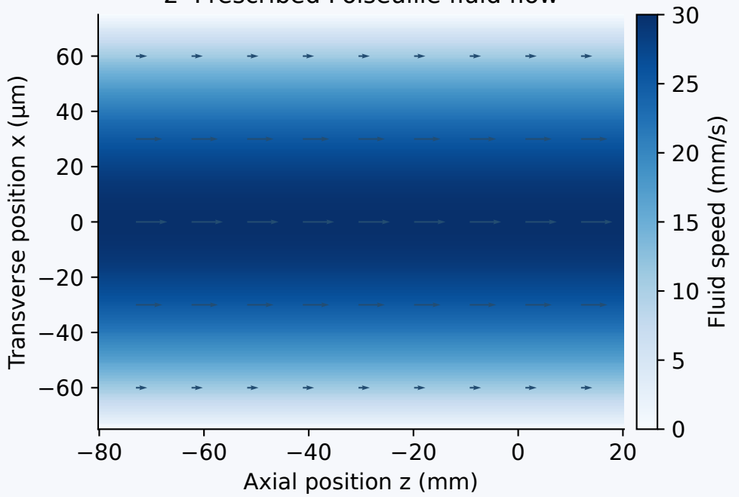
1a Prescribed magnetic amplitudes



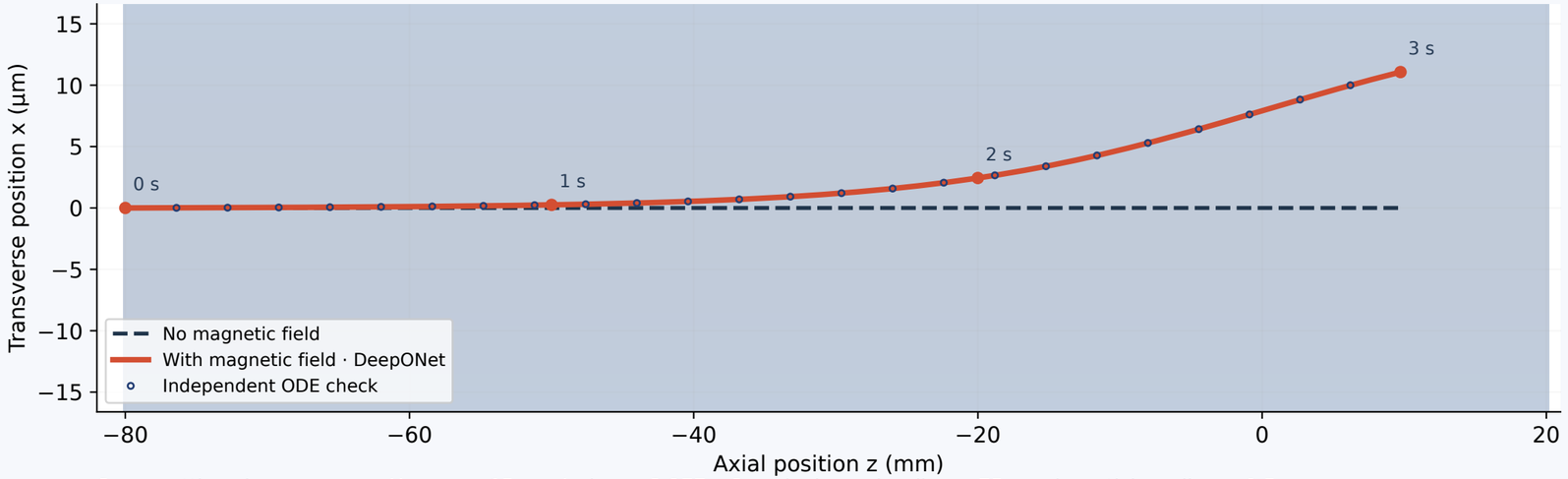
1b Magnetic field at t = 0.00 s



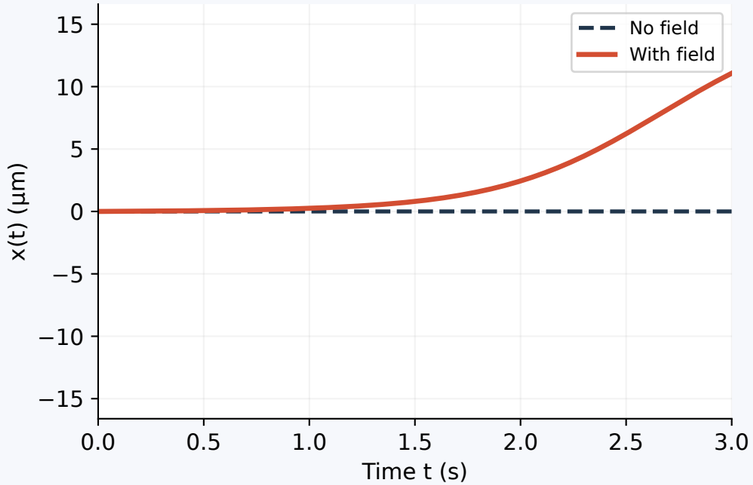
2 Prescribed Poiseuille fluid flow



3a Same release and flow: trajectories with / without magnetic forcing



3b Transverse particle displacement



Same settings in every case: Umean = 15 mm/s | η = 3.277 mPa·s | channel radius = 75 μm | particle radius = 0.5 μm

Time-varying amplitudes; fixed spatial source geometry. Overdamped induced-dipole model. Axial and transverse axes use different units.

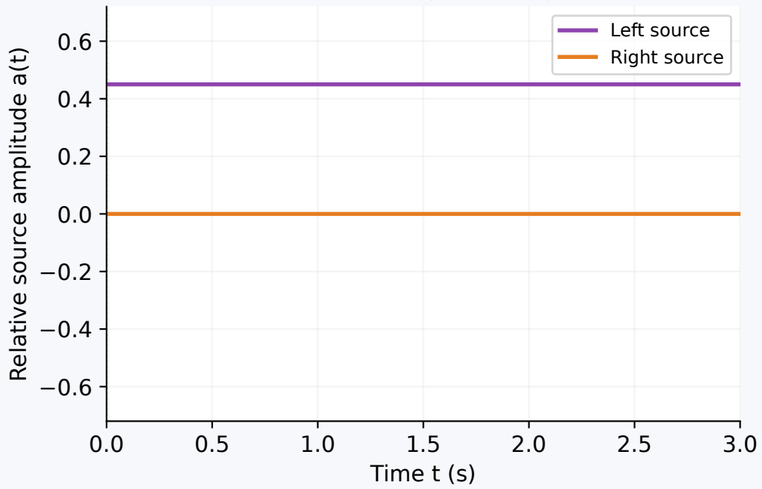
Independent max error: x 0.007 μm · z 1.21 μm

MAGNETIC INPUT → FLUID FLOW → PARTICLE RESPONSE

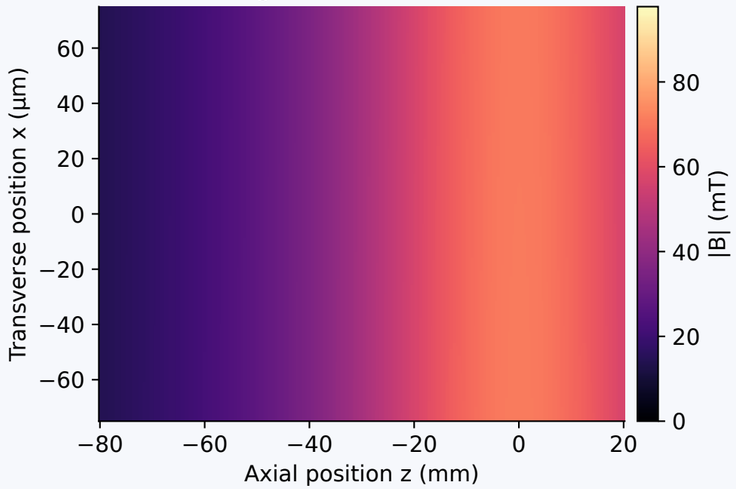
Constant left source

$a_L = 0.45, \quad a_R = 0$

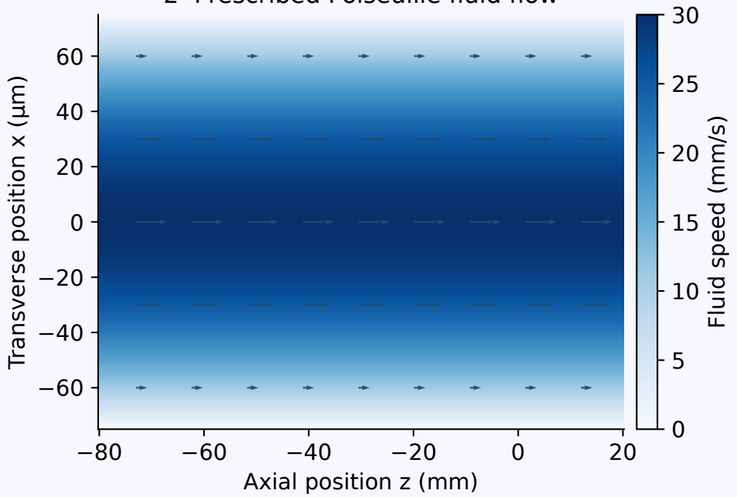
1a Prescribed magnetic amplitudes



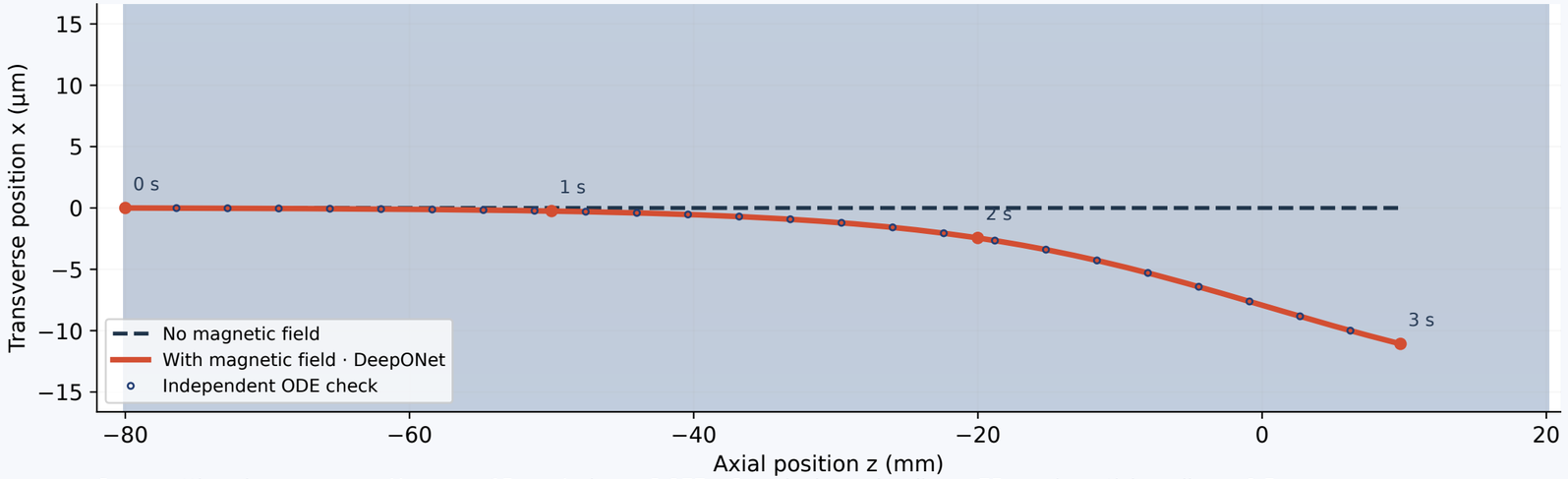
1b Magnetic field at t = 0.00 s



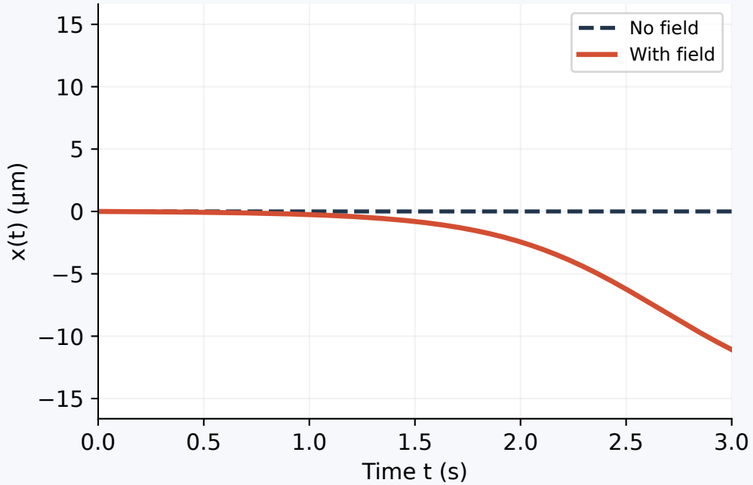
2 Prescribed Poiseuille fluid flow



3a Same release and flow: trajectories with / without magnetic forcing



3b Transverse particle displacement



Same settings in every case: Umean = 15 mm/s | η = 3.277 mPa·s | channel radius = 75 μm | particle radius = 0.5 μm

Time-varying amplitudes; fixed spatial source geometry. Overdamped induced-dipole model. Axial and transverse axes use different units.

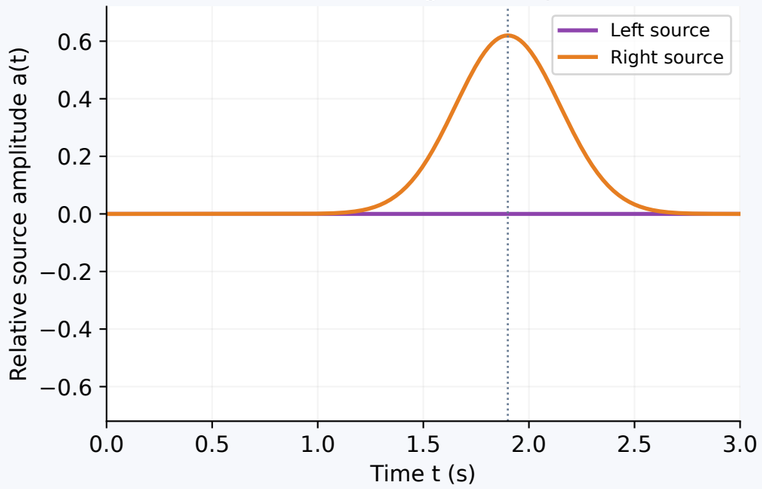
Independent max error: x 0.001 μm · z 0.94 μm

MAGNETIC INPUT → FLUID FLOW → PARTICLE RESPONSE

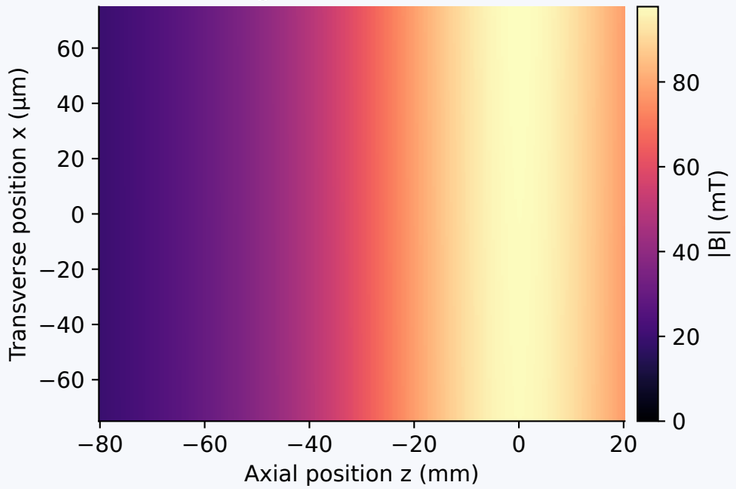
Gaussian pulse on the right

$a_L = 0, \quad a_R = 0.62e^{-((t-1.9)/0.35)^2}$

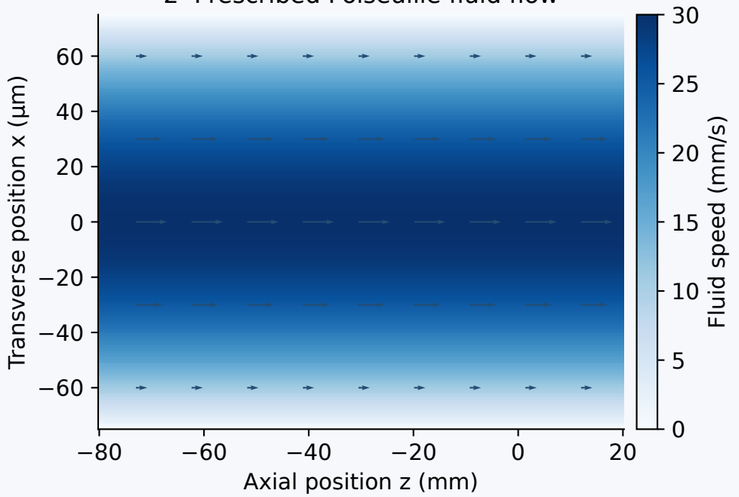
1a Prescribed magnetic amplitudes



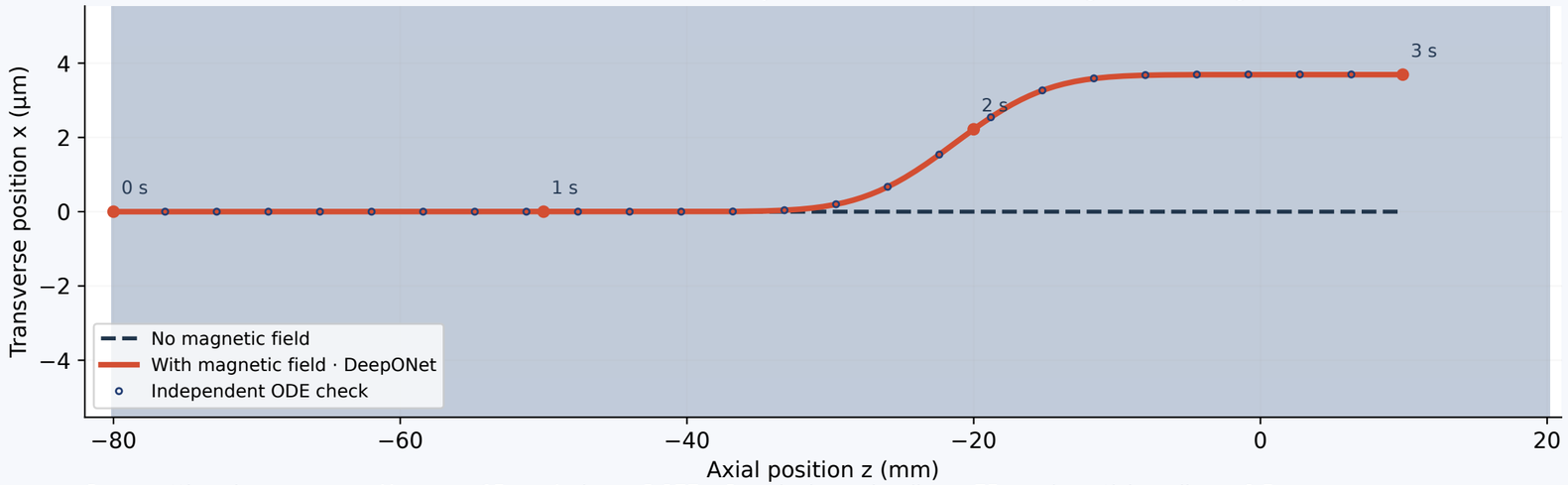
1b Magnetic field at t = 1.90 s



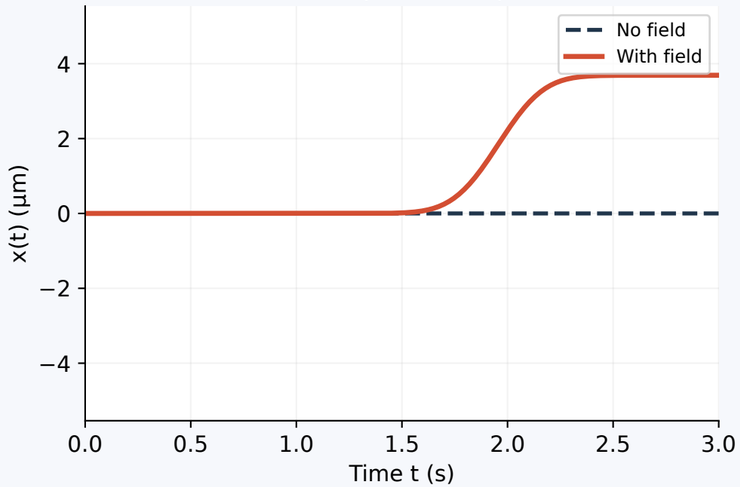
2 Prescribed Poiseuille fluid flow



3a Same release and flow: trajectories with / without magnetic forcing



3b Transverse particle displacement



Same settings in every case: Umean = 15 mm/s | η = 3.277 mPa·s | channel radius = 75 μm | particle radius = 0.5 μm

Time-varying amplitudes; fixed spatial source geometry. Overdamped induced-dipole model. Axial and transverse axes use different units.

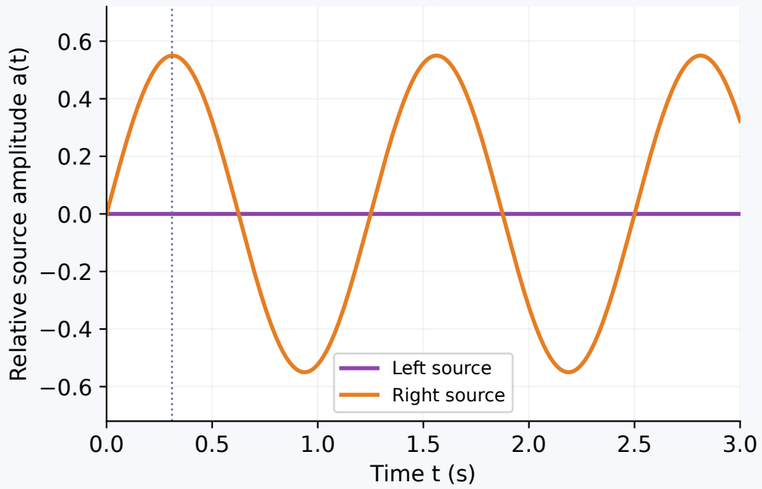
Independent max error: x 0.005 μm · z 0.16 μm

MAGNETIC INPUT → FLUID FLOW → PARTICLE RESPONSE

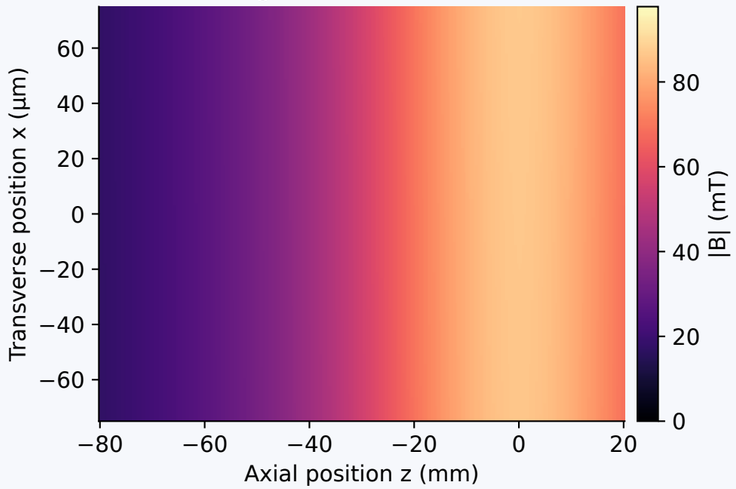
Sinusoidal right source

$a_L = 0, \quad a_R = 0.55\sin(2\pi\,0.8t)$

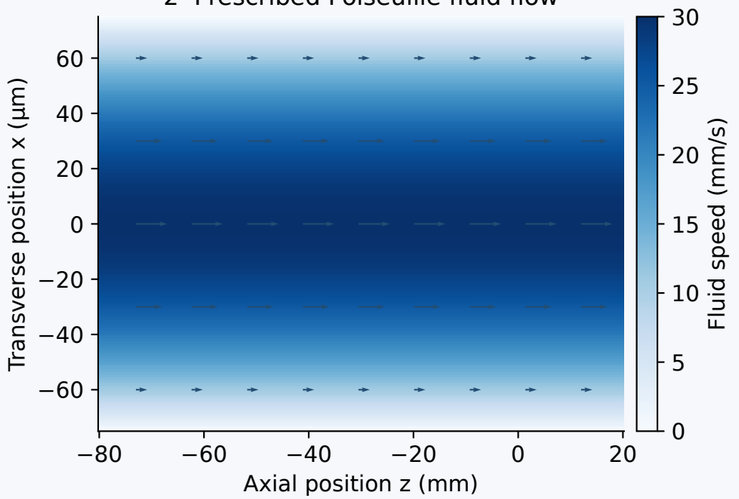
1a Prescribed magnetic amplitudes



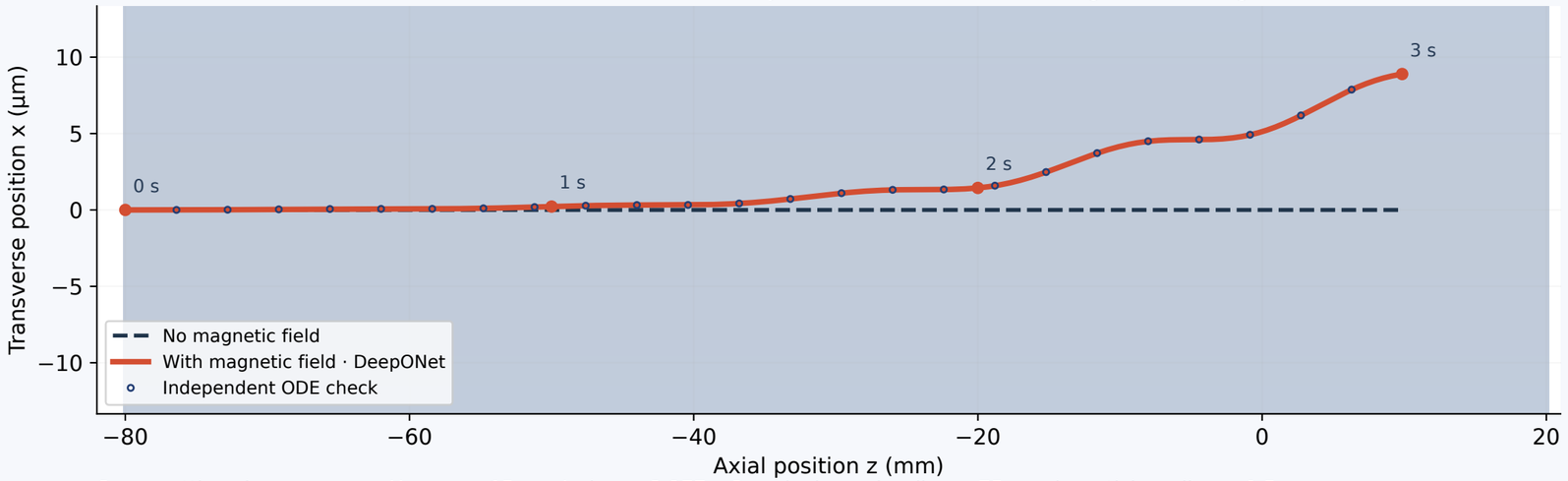
1b Magnetic field at t = 0.31 s



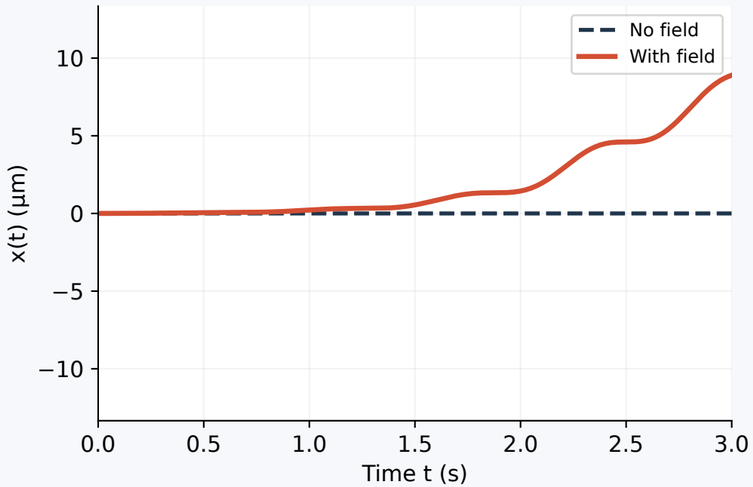
2 Prescribed Poiseuille fluid flow



3a Same release and flow: trajectories with / without magnetic forcing



3b Transverse particle displacement



Same settings in every case: Umean = 15 mm/s | η = 3.277 mPa·s | channel radius = 75 μm | particle radius = 0.5 μm

Time-varying amplitudes; fixed spatial source geometry. Overdamped induced-dipole model. Axial and transverse axes use different units.

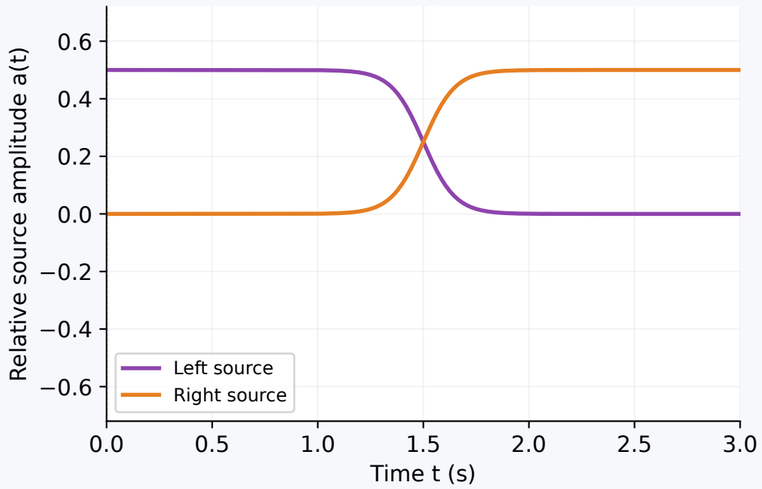
Independent max error: x 0.008 μm · z 0.57 μm

MAGNETIC INPUT → FLUID FLOW → PARTICLE RESPONSE

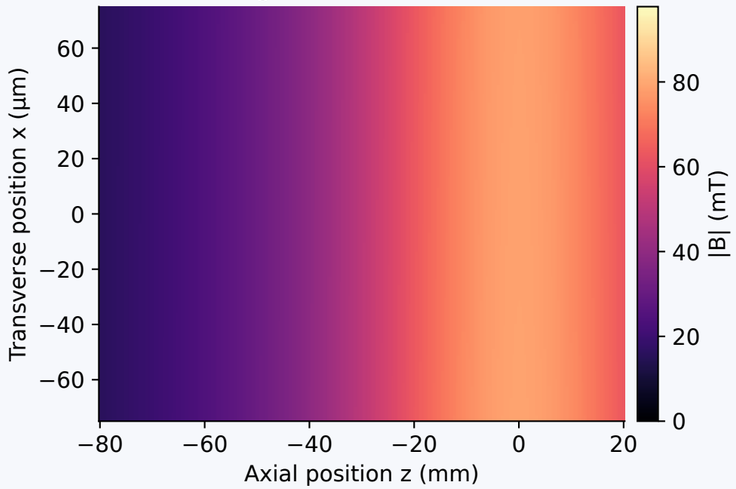
Smooth switch: left to right

$a_{L,R} = 0.25[1 \mp \tanh((t - 1.5)/0.15)]$

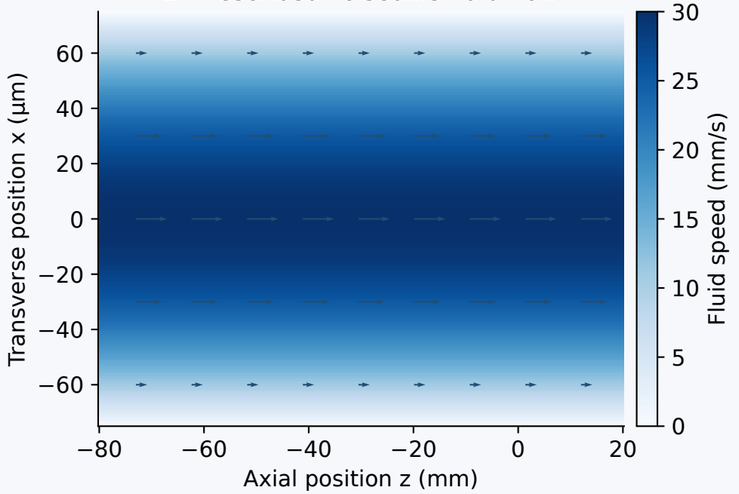
1a Prescribed magnetic amplitudes



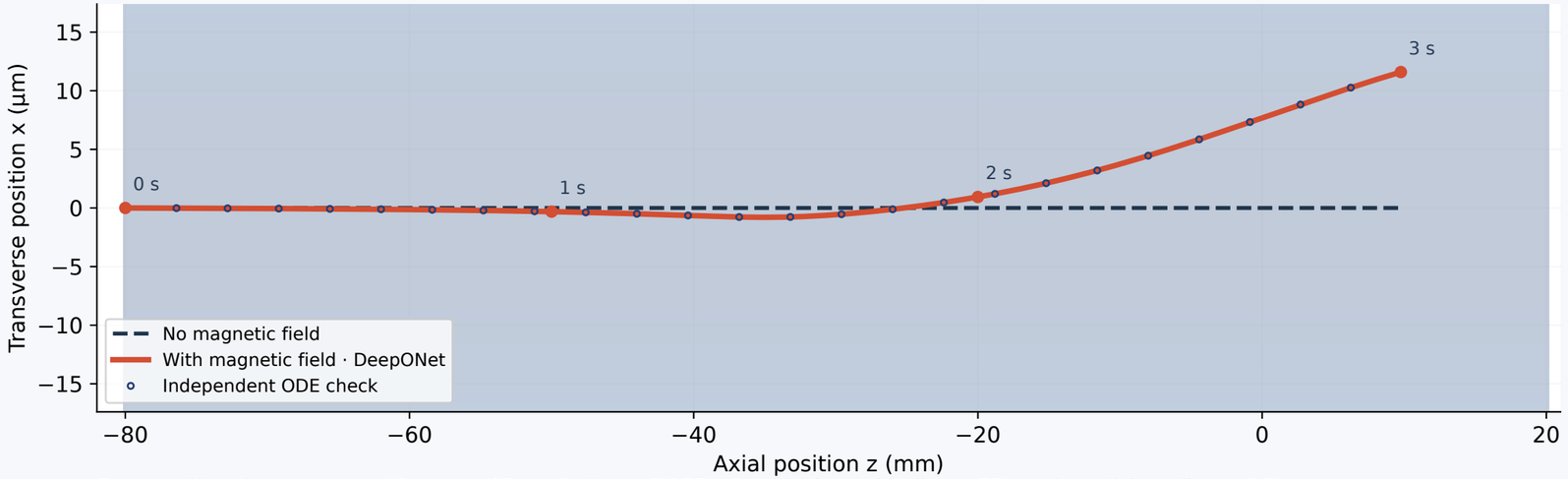
1b Magnetic field at t = 0.00 s



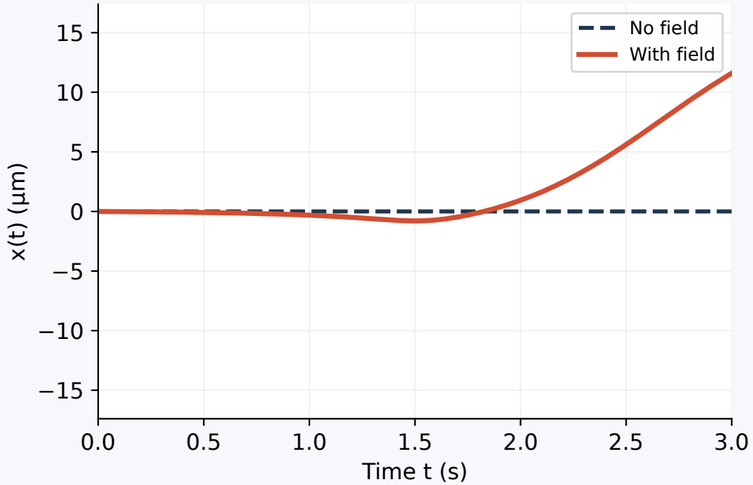
2 Prescribed Poiseuille fluid flow



3a Same release and flow: trajectories with / without magnetic forcing



3b Transverse particle displacement



Same settings in every case: Umean = 15 mm/s | η = 3.277 mPa·s | channel radius = 75 μm | particle radius = 0.5 μm

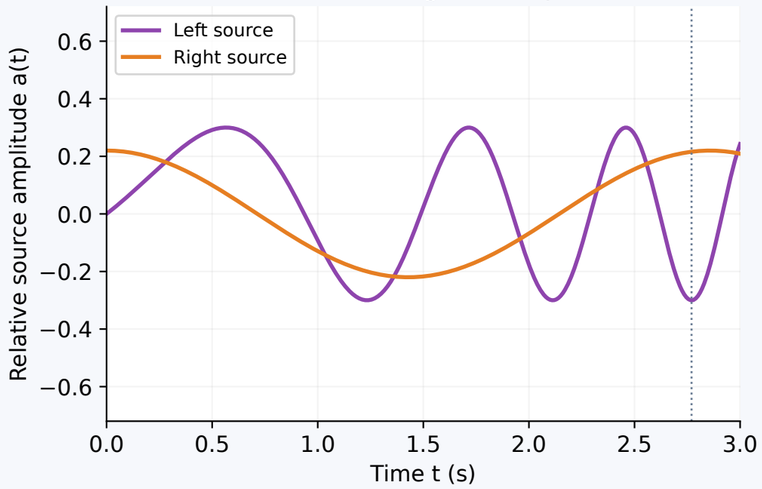
Time-varying amplitudes; fixed spatial source geometry. Overdamped induced-dipole model. Axial and transverse axes use different units.

Independent max error: x 0.004 μm · z 0.88 μm

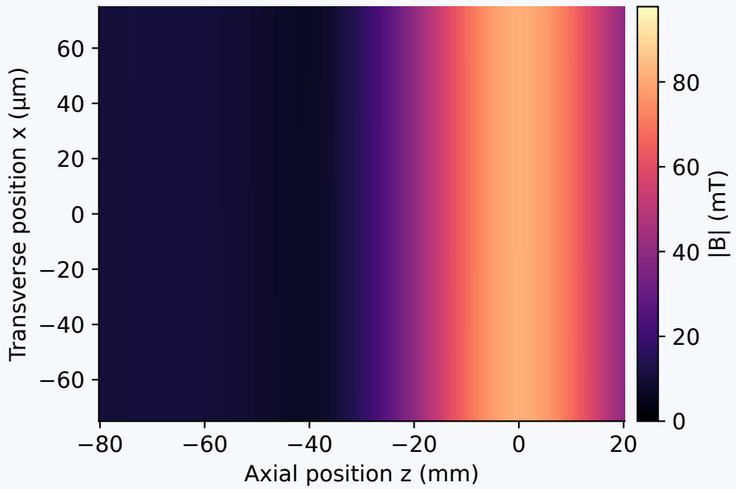
Two-source chirp and cosine

$a_L = 0.30\sin(2\pi(0.3t + 0.25t^2)), \quad a_R = 0.22\cos(2\pi\,0.35t)$

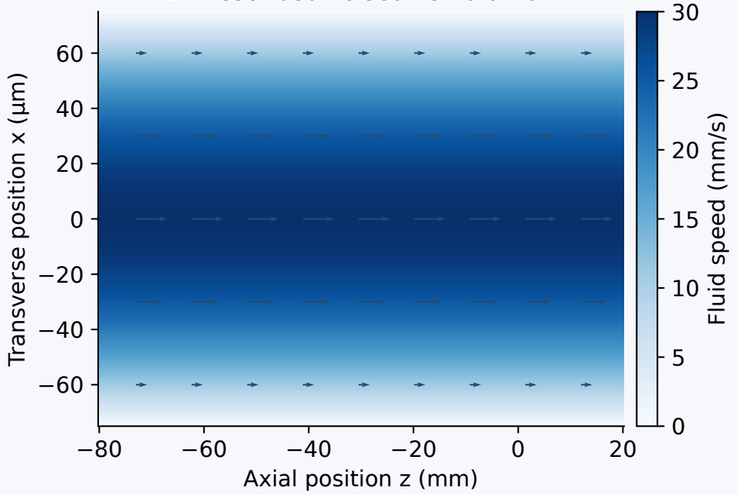
1a Prescribed magnetic amplitudes



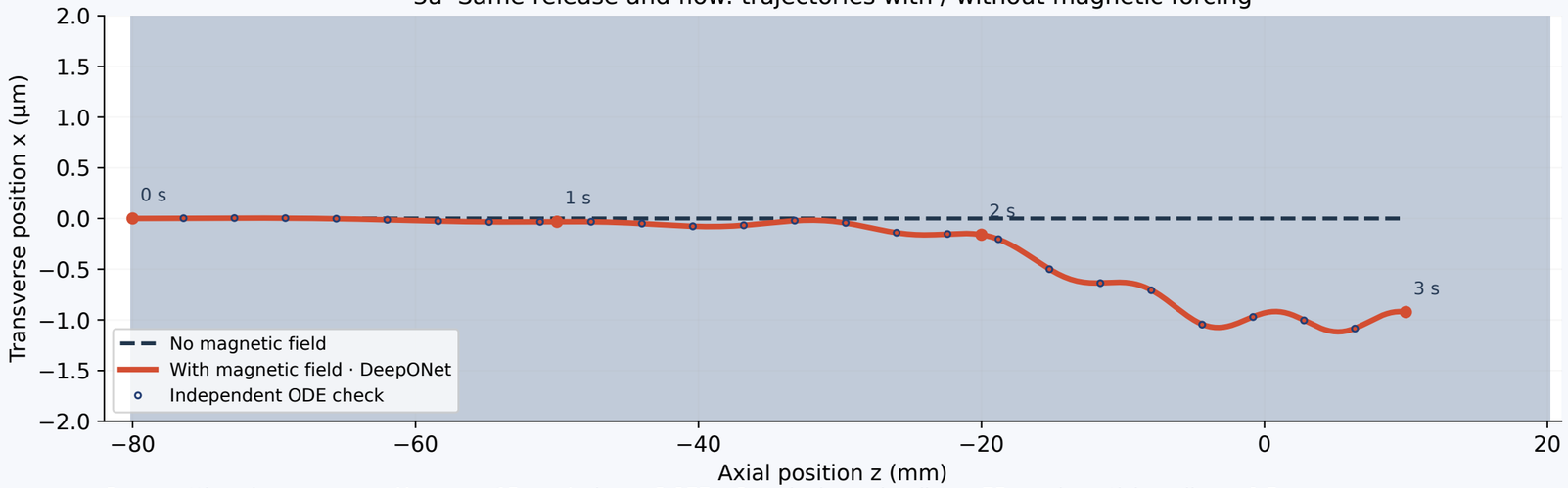
1b Magnetic field at t = 2.77 s



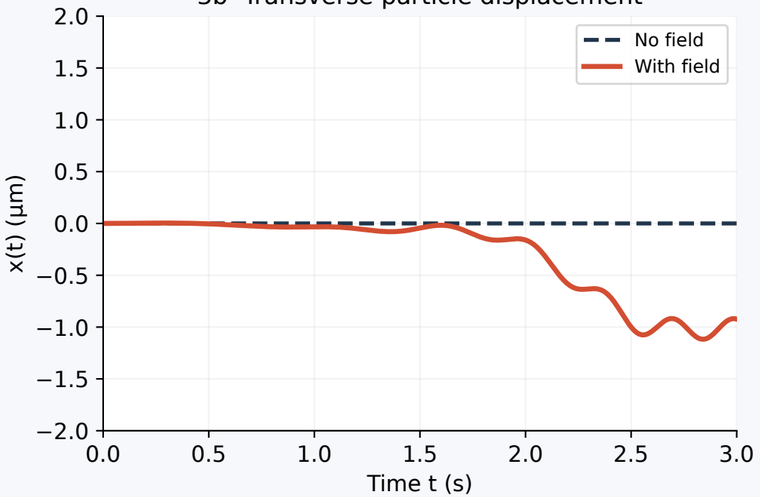
2 Prescribed Poiseuille fluid flow



3a Same release and flow: trajectories with / without magnetic forcing



3b Transverse particle displacement



Same settings in every case: Umean = 15 mm/s | η = 3.277 mPa·s | channel radius = 75 µm | particle radius = 0.5 µm

Time-varying amplitudes; fixed spatial source geometry. Overdamped induced-dipole model. Axial and transverse axes use different units.

Independent max error: x 0.003 µm · z 0.61 µm